



REAL LIFE ANECDOTE: EMPOWERING YOU TO CREATE THE BOX PLOT

Ram: “Welcome back, everyone! This is not just another lecture; it's a call to action, a chance to apply what you've learned, and make a real-world impact.

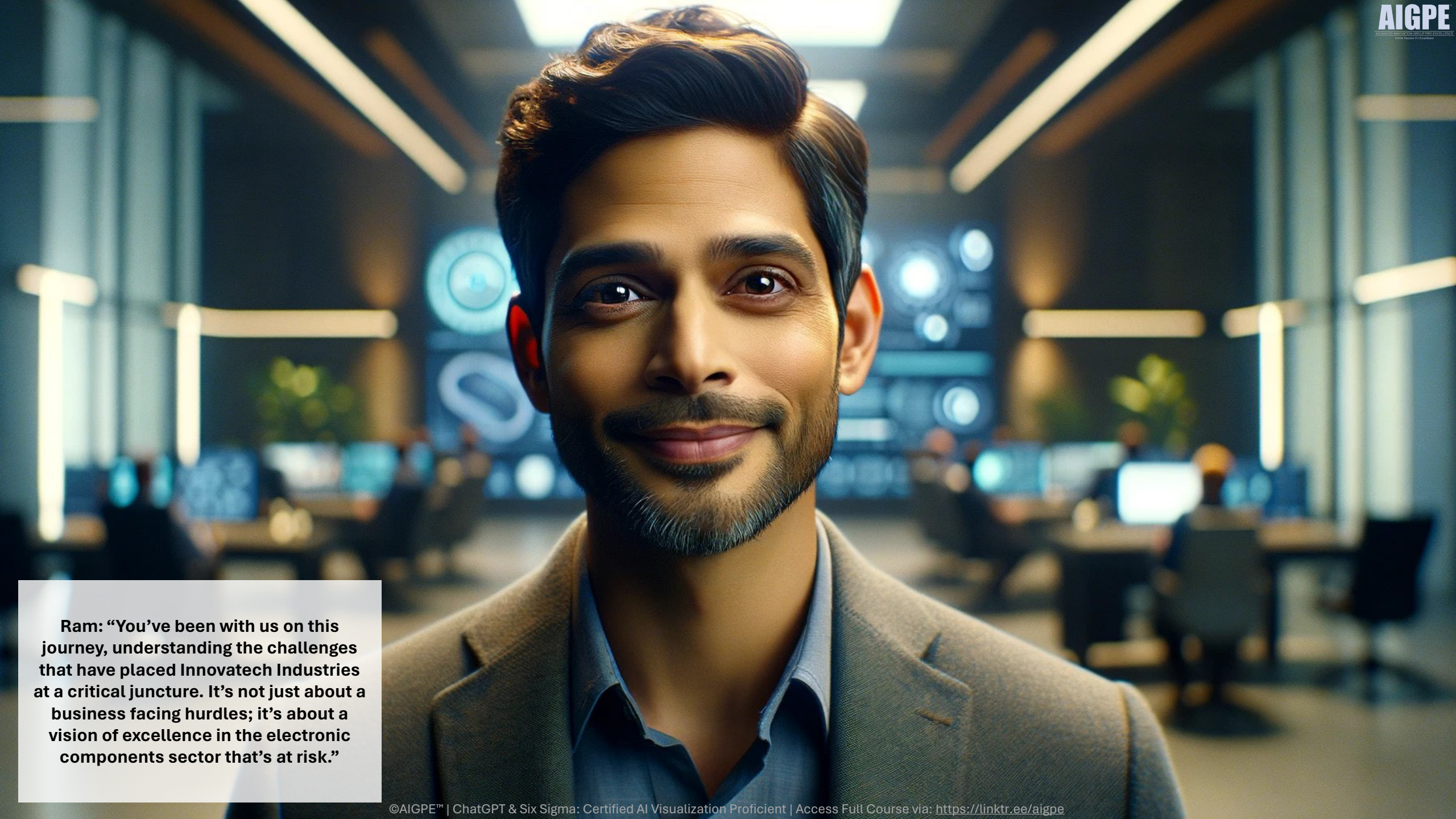
You're not just students; you're Innovatech's newest problem-solvers!”

Ram: “Let's kick things off with the data we've got. On the screen you will see a table neatly organized into five columns.”

BUSINESS SCENARIO – REVIEWING THE DATA

	A	B	C	D	E
1	Line A	Line B	Line C	Line D	Line E
2	1.44	4.88	0.27	3.66	4.57
3	2.88	4.10	1.66	1.19	2.34
4	2.33	4.76	0.92	0.84	2.65
5	2.00	4.58	1.37	2.11	5.59
6	0.89	3.39	2.29	3.95	4.43
7	0.89	4.69	0.77	1.20	2.04
8	0.65	1.35	1.14	2.79	2.41
9	2.67	1.78	1.94	3.12	4.65
10	2.00	1.18	0.73	1.18	2.02
11	2.27	2.30	0.38	2.99	2.64
12	0.55	2.55	0.87	1.66	4.19
13	2.92	2.09	0.57	2.64	4.77
14	2.58	4.31	2.34	2.64	4.61
15	1.03	2.43	2.06	2.28	2.90
16	0.95	2.12	1.66	0.63	4.85
17	0.96	3.17	2.20	3.39	2.95
18	1.26	1.56	2.05	1.49	3.30
19	1.81	4.21	0.63	0.99	4.99
20	1.58	1.30	2.25	0.45	4.60

File Name: **1 Box Plot Data.csv**

A man with dark hair and a beard, wearing a grey suit jacket over a blue shirt, is smiling slightly. He is in a modern, futuristic office with glowing blue and white lights, blurred desks, and people in the background.

Ram: “You’ve been with us on this journey, understanding the challenges that have placed Innovatech Industries at a critical juncture. It’s not just about a business facing hurdles; it’s about a vision of excellence in the electronic components sector that’s at risk.”

BUSINESS SCENARIO

BUSINESS SITUATION BACKGROUND

INNOVATECH INDUSTRIES, RENOWNED FOR SPEARHEADING INNOVATION IN THE ELECTRONICS SECTOR, HAS ENCOUNTERED A SIGNIFICANT HURDLE THAT JEOPARDIZES ITS MARKET DOMINANCE AND REPUTATION FOR QUALITY. DESPITE ITS ROBUST INNOVATION PIPELINE, THE COMPANY HAS SEEN AN UPTICK IN CUSTOMER COMPLAINTS AND RETURNS, POINTING TO AN UNDERLYING ISSUE OF INCONSISTENT PRODUCT QUALITY ACROSS ITS MANUFACTURING LINES. AMIT, THE VISIONARY CEO OF INNOVATECH, FACES THE CHALLENGE HEAD-ON, RECOGNIZING THAT THE COMPANY'S FUTURE HINGES ON ITS ABILITY TO DELIVER CONSISTENTLY HIGH-QUALITY PRODUCTS.

PROBLEM STATEMENT

THE CORE ISSUE AT INNOVATECH LIES IN THE VARIABILITY OF PRODUCT QUALITY ACROSS ITS VARIOUS PRODUCTION LINES. THIS INCONSISTENCY HAS LED TO AN INCREASE IN DEFECT RATES, DIRECTLY IMPACTING CUSTOMER SATISFACTION AND LOYALTY. AMIT IS AWARE THAT WITHOUT DETAILED INSIGHTS INTO THE PERFORMANCE OF EACH PRODUCTION LINE, DEVISING AN EFFECTIVE SOLUTION REMAINS A CHALLENGE. THE COMPANY URGENTLY NEEDS TO PINPOINT WHICH PRODUCTION LINES ARE FALLING SHORT OF QUALITY STANDARDS TO IMPLEMENT TARGETED IMPROVEMENTS, THEREBY ENSURING PRODUCT RELIABILITY AND UPHOLDING ITS COMMITMENT TO EXCELLENCE.

TASK: CREATE A BOX PLOT

YOUR TASK IS TO CREATE A BOX PLOT TO ANALYZE THE DEFECT RATES ACROSS INNOVATECH'S PRODUCTION LINES, PINPOINTING VARIABILITY AND OUTLIERS. THIS FOCUSED ANALYSIS WILL HELP IDENTIFY WHICH LINES NEED URGENT QUALITY IMPROVEMENTS, GUIDING STRATEGIC DECISIONS TO ENHANCE PRODUCT CONSISTENCY AND REBUILD CUSTOMER

PLEASE DOWNLOAD THE DATA IN THE .CSV FORMAT FOR THIS EXERCISE FROM THE RESOURCES SECTION OF THIS LECTURE

DATA

AI PROMPTS

**TEMPLATE FOR
 UPDATING
 RECOMMENDATIONS**

**SHARE YOUR FINDINGS
 ON THE Q&A
 DISCUSSION BOARD**

Ram: “I want to thank you in advance for joining us in this critical endeavor. Your insights and recommendations could very well be the key to turning Innovatech around. Let's make a difference together. Let's save Innovatech. Let's innovate.”

AIGPETM

ADVANCED INNOVATION GROUP PRO EXCELLENCE

Drive Success To Excellence